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Project Proposal: **Political Bias Comparison**

Reading the news lately feels like reading two completely different realities. This project is about building an NLP pipeline to measure exactly how different media outlets “spin” the same story. Using a labeled dataset like **AllSides** or **BASIL**, to run a **Named Entity Recognition** (NER) model to pull out the main people and places in an article (**BERT-based NER)**.

Then instead of just summarizing the whole text, the pipeline will extract the specific sentences surrounding those entities and feed them into a local transformer model. By running an **Aspect-Based Sentiment Analysis**. The goal is to explore how effectively a chained NLP pipeline can isolate and quantify media bias toward specific subjects from different sides of the political spectrum.

To evaluate the pipeline’s effectiveness, I’ll compare its sentiment scores against the expert annotations in the dataset or a neutral baseline source. Also include an error analysis to catch the NLP errors, like tracking pronouns back to the right person. IE when a news article uses different names for the same politician.

The final deliverable will be an interactive Streamlit dashboard where users can compare opposing articles side-by-side and visualize their respective sentiment scores. This work will provide insight into automation and the practical deployment of LLMs for multi-step text analysis.